PRADEEP MALLAMPALLI

DATA SCIENTIST

CONTACT

+1 704-649-1042

pradeep.dst1@gmail.com

Charlotte, NC

linkedin.com/in/pradeepmallampalli

SKILLS

AI/ML & Deep Learning:

- TensorFlow, Neural Networks, RNN, CNN

Generative AI:

- LLMs, RAG, LangChain, OpenAI, Prompt Engineering

Data Engineering:

- ETL, PySpark, Data Pipelines, Hadoop, GraphDB, SQL

Analytics & Visualization:

- EDA, Tableau, Statistical Analysis, Dashboarding

Programming & Development:

- Python, SQL, GSQL, Git, Jira

NLP & Text Processing:

- Classification, Semantic Analysis, Text Mining

Cross-Platform:

- Windows, MacOS, Linux

PROFILE

Accomplished Data Scientist with 13+ years of experience, including 7+ years focused on AI/ML and Generative AI solutions across industries such as Banking, Healthcare, and Oil & Gas. Proven expertise in deploying advanced frameworks like RAG (Retrieval-Augmented Generation), LangChain, and multimodal AI systems. Recently led high-impact Gen AI projects, including chatbots, document Q&A systems, and large-scale information extraction pipelines. Certified Generative AI Professional (Infosys, 2024) with hands-on experience in LLMs, OpenAI, and cutting-edge NLP techniques.

GEN AI PROJECTS

Multimodal RAG for Document Q&A System — 2025

- Developed a sophisticated document analysis system capable of understanding and reasoning across both textual and visual elements within complex documents
- Created custom preprocessing pipelines for various document formats to maintain structural integrity
- Engineered specialized retrievers for different content types (text, tables, images) with a unified ranking mechanism

AI-Based Internal Organizational Chatbot — 2024

- Built a Retrieval-Augmented Generation (RAG) system using LangChain to extract insights from massive enterprise datasets
- Implemented multi-agent workflows, allowing the AI to autonomously refine its answers.
- Created a dynamic chunking system that adapts to document structure and content type for improved context preservation.

WORK EXPERIENCE

Trade Analytics - Anomaly Detection — Apr-2023- Present

Data Scientist - AI/ML | Bank of America - Infosys

- Developed an advanced anomaly detection framework leveraging statistical modeling and AI, significantly reducing false alerts in trade monitoring.
- Conducted EDA on large-scale trade data, identifying key statistical patterns for anomaly detection.
- Utilized big data technologies (PySpark, SQL, Python) to efficiently process and analyze terabytes of trade data
- Achieved 70% reduction in false positives compared to previous rule-based methods.

EDUCATION

Nagarjuna University

- Bachelors Degree (MECH)

CERTIFICATIONS

IBM Data Science Professional Certificate

- Coursera (2021)

Infosys Certified Generative AI Professional

- Infosys -2024

Disaster Recovery Acceleration

Jul-2020 - Mar 2023

Data Scientist - AI/ML | Bank of America - Infosys

- DRA (Disaster Recovery Acceleration) Program responsible for identification of Data corruption faster and stop propagation of corruption to downstream systems
- Created machine learning model to categorize AutoSys jobs into data movement and non-data movement classifications using NLP, leveraging job names and descriptions
- Building and integration entire pipeline with blend of technologies comprising ML, Pyspark , Python, Graph DB, SQL, ETL, Hive and Oracle

Image Segmentation Using CNN

Oct-2019 - Jun 2020

Data Scientist - AI/ML | GE - Infosys

- Designed a CNN-based model for medical image segmentation, enhancing diagnosis and treatment planning.
- Achieved high segmentation accuracy for anatomical structures using transfer learning techniques.

Credit Default Identification

Feb-2018 - Sep 2019

Data Scientist - AI/ML | Commonwealth Bank - Infosys

- Develop and maintain credit risk models for PPNR and credit risk models leveraging upon Python, SQL, Tableau and other ETL Technologies
- Perform EDA using Python libraries for generating various graphs and charts for analyzing and implementing Machine Learning

Lithology Prediction

Aug-2017 - Jan 2018

ML Engineer | BHP - Infosys

- Analyze large datasets and Developing ML Models and presenting project findings
- Achieved high segmentation accuracy for anatomical structures using transfer learning techniques.

Aerospace Structural Analyst

Apr-2011 - Jul 2017

Design Engineer | Boeing - Infosys

- Leverage CAD and finite element analysis FEA tools to simulate and test the behavior of the structures under various conditions such as aerodynamic loads, thermal stresses, and vibrations.
- Performs design and stress analysis of structure using classical hand methods and assessing the strength, stability & performance of structures